

Questions		Answers	Marks
4	(a)	There is no further gain in KE as the acceleration of the metal sphere reaches zero, leading to a constant velocity.	[B1]
		The acceleration reaches zero as the viscous (resistive) force increases with velocity of the sphere, until the viscous force equals to the weight of the sphere.	[B1]
	(b)	According to Newton's second law, this force results in an acceleration a where	
		$F = ma$	[B1]
		The work done by the constant force F is	
		$W = F.s = mas$	[B1]
		But $v^2 = u^2 + 2as$	
		$\rightarrow as = \frac{1}{2}(v^2 - u^2)$	
		$\therefore W = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$	
		= final kinetic energy – initial kinetic energy	[B1]
		= ΔE_k	
		If the object start from rest ($\frac{1}{2}mu^2 = 0$),	
		$\therefore W = \frac{1}{2}mv^2 \text{ (derived)}$	[A0]
		Constant acceleration is necessary for the derivation	[B1]

